

25-09-2026

PRACTICAL - 9

SUBNETTING

AIM:-

Implementation of subnetting in CISCO PACKET TRACER simulator.

INTRODUCTION:-

Subnetting is the process of dividing a large IP network into smaller logical networks called subnets. Classless subnetting allows us to use subnet masks other than the default class-based masks, making IP addresses more efficient and allowing multiple networks to be created.

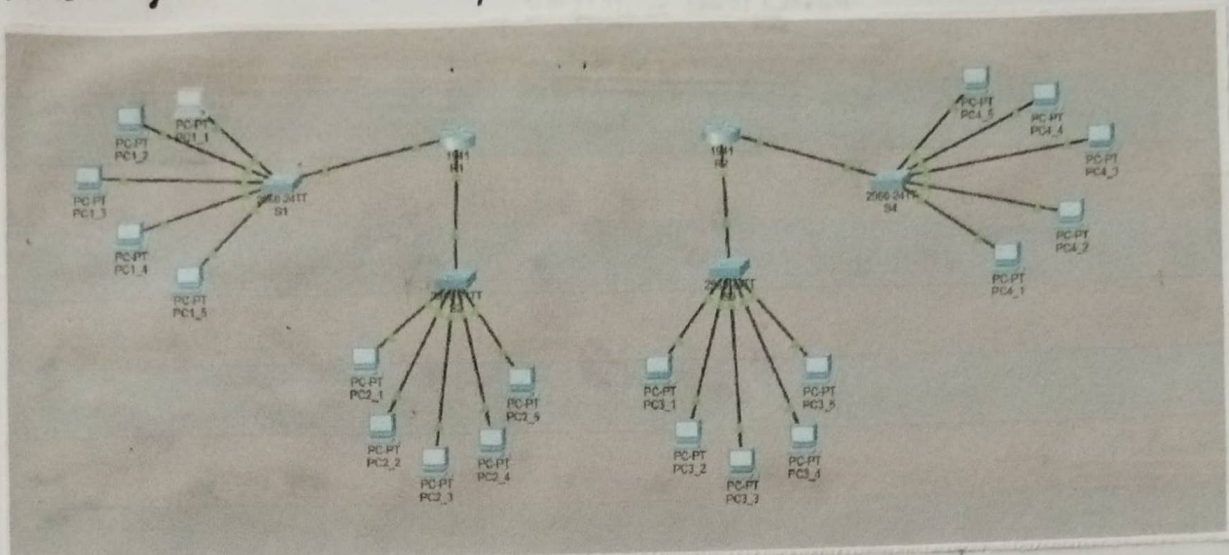
Here the new $192.168.1.0/24$ is divided into smaller 127 subnets.

A 127 subnet provides 30 usable host addresses per subnet.

PROCEDURE :-

- 1) Open Cisco Packet Tracer.
- 2) Add required routers, switches and PCs.
- 3) Connect the devices using appropriate cables.
- 4) Configure the routers with the required IP addresses and subnet masks.
- 5) Configure the router interfaces using the CLI:-
 - enable
 - configure terminal
 - select the required Fast Ethernet / Gigabit Ethernet Interface.
 - Assign the IP address and subnet mask.
 - Use no shutdown to activate the interface.
- 6) Configure the switch ports in access mode where required.
- 7) Assign static IP addresses, subnet masks and default gateways to the PCs.
- 8) Use the 127 subnet mask: $255.255.255.224$.
- 9) Configure the four subnets:-
 - $192.168.1.0/27$
 - $192.168.2.0/27$
 - $192.168.3.0/27$
 - $192.168.4.0/27$
- 10) Verify the router interfaces and PCs belong to their respective routers.
- 11) Open command prompt on the PCs.
- 12) Use the ping command to test connectivity between PC and respective routers.

Subnetting Network topology:-



Ping used to verify communication between PCs in different subnets.

```
PC1_1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1
Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.1.12
Pinging 192.168.1.12 with 32 bytes of data:
Reply from 192.168.1.12: bytes=32 time=1ms TTL=128
Reply from 192.168.1.12: bytes=32 time=1ms TTL=128
Reply from 192.168.1.12: bytes=32 time=1ms TTL=128
Reply from 192.168.1.12: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.1.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.1.11
Pinging 192.168.1.11 with 32 bytes of data:
Reply from 192.168.1.11: bytes=32 time=1ms TTL=128
Reply from 192.168.1.11: bytes=32 time=1ms TTL=128
Reply from 192.168.1.11: bytes=32 time=1ms TTL=128
Reply from 192.168.1.11: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 1ms
C:\>ping 192.168.2.11
Pinging 192.168.2.11 with 32 bytes of data:
Request timed out.
Reply from 192.168.2.11: bytes=32 time=1ms TTL=127
Reply from 192.168.2.11: bytes=32 time=1ms TTL=127
Reply from 192.168.2.11: bytes=32 time=1ms TTL=127
Reply from 192.168.2.11: bytes=32 time=1ms TTL=127
Ping statistics for 192.168.2.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 8ms, Average = 2ms
C:\>
```

```
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.2.11
Pinging 192.168.2.11 with 32 bytes of data:
Reply from 192.168.2.11: bytes=32 time=1ms TTL=127
Reply from 192.168.2.11: bytes=32 time=1ms TTL=127
Reply from 192.168.2.11: bytes=32 time=1ms TTL=127
Reply from 192.168.2.11: bytes=32 time=1ms TTL=127
Ping statistics for 192.168.2.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 8ms, Average = 2ms
C:\>
```

Slk

RESULT:-

Thus, subnetting was successfully implemented using CISCO PACKET TRACER.